



School of Pharmacy, Fudan University



Proteases play a crucial role in essential physiological processes and are recognized as therapeutic targets and diagnostic biomarkers for different pathologies. Here, we developed a new class of Tetracysteine-based Protease Activity Reporters (TePARs) by sandwiching peptides containing protease cleavage sites in between the NanoLuc protein and tetracysteine motif. TePARs used bioluminescence resonance energy transfer (BRET) to shift emitted light from green or red to blue upon proteolysis, allowing detection with either a plate reader or a simple smartphone camera. With further optimization, TePAR 2.0 utilized tandem tetracysteine peptides to achieve unprecedented BRET efficiency (~90%), large dynamic range (>40-fold), and a picomolar limit of detection (LoD). Moreover, TePARs feature a modular design that enables the generation of a wide range of protease sensors by simply swapping peptide substrates for different proteases while maintaining exceptional performance. We used TePAR to quantify active proteases at physiologically relevant concentrations in challenging biological samples, yielding results consistent with those from standard testing methods. Notably, TePARs enable the quantification of MMP-9 plasma levels of colorectal cancer patients at the point of care. TePAR assay distinguished patients with colorectal cancer, including those with early-stage adenomatous polyps, from healthy controls with accuracies of over 90%, thus expanding their utility for point-of-care diagnostics. Furthermore, we also demonstrate that TePAR enables camera-based drug screening and characterization without the need for specialized instruments.

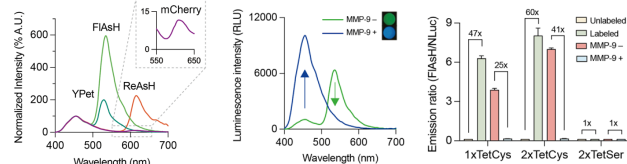
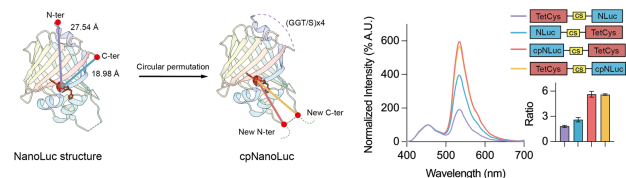
Tetracycline-based Protease Activity Reporter (TEPAR)

The diagram illustrates the TEPAR system. On the left, a NanoLuc (NLuc) protein (blue circle) is fused to a tetracycline tag (red hexagons) via a cleavage site (yellow circles). A bis-arsenical dye (red hexagons) is attached to the tetracycline tag. The NLuc protein is shown in a blue circle. The tetracycline tag is shown as a chain of red hexagons. The cleavage site is shown as a chain of yellow circles. A protease (grey shape) is shown cleaving the tetracycline tag. The resulting NLuc protein is shown in a blue circle, and the tetracycline tag is shown as a chain of red hexagons. The bis-arsenical dye is shown as a chain of red hexagons. The NLuc protein is shown in a blue circle. The tetracycline tag is shown as a chain of red hexagons. The cleavage site is shown as a chain of yellow circles. A protease (grey shape) is shown cleaving the tetracycline tag. The resulting NLuc protein is shown in a blue circle, and the tetracycline tag is shown as a chain of red hexagons. The bis-arsenical dye is shown as a chain of red hexagons.

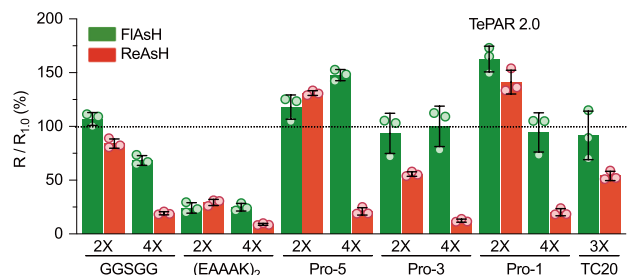
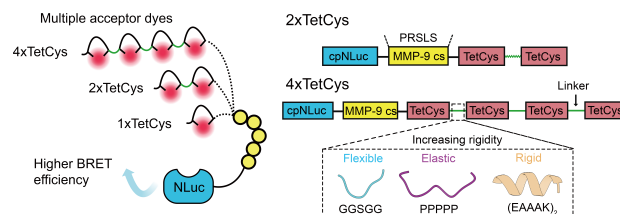
Chemical structures of the dyes are shown on the right:

- FIAsH-EDT₃ (top right)
- ReAsH-EDT₃ (bottom right)

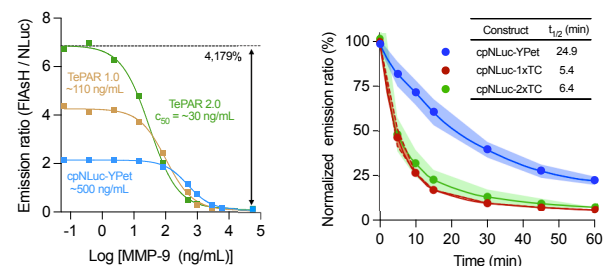
1. Development of a chemigenetic BRET sensor for MMP-9



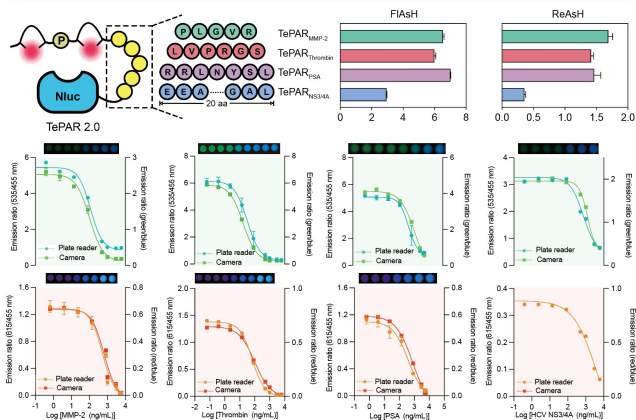
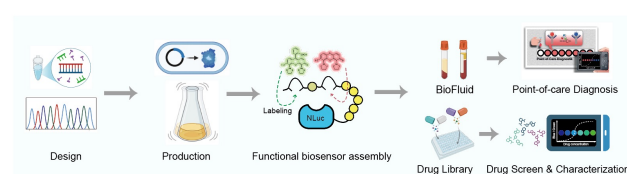
2. Sensor protein optimization using tandem-tetracysteine tags



3. Characterization of MMP-9 BRET sensor



4. Modularity in the analysis of different proteases



5. Application in protease activity quantification in human plasma

